

THE APPLICATION OF ELECTRONIC COMPUTERS TO FACTOR ANALYSIS¹

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IN one sense, it is appropriate that I should go first on this symposium. For my topic is electronic computers in factor analysis—and factor analysis may properly be said to be that technique in psychology for which computers were first used and are unquestionably the most used today. Thus, relatively, I shall be talking about a more traditional, better established, and consequently perhaps more stodgy and less exciting application of computers to psychological problems.

Let me warn you about how I am going to talk today. I have not conducted a survey of available computer programs for factor analytic computations, nor have I done an analysis of the problems of the application of computers to factor analysis in any way that could be considered scientific. I am saying that I shall ask you to listen to my opinions about the applications of computers to factor analysis and only hope that these opinions and anecdotes make some sense. My qualification for presuming to ask you to listen to these ruminations is purely quantitative: I imagine that I am guilty of having carried out more factor analytic calculations of the theoretical sort on electronic computers than anyone walking the earth.

Practical Application

While almost my entire paper is going to be about the implications of computers for the theory of factor analysis, allow me to devote a

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I am most indebted to Professor N. L. Gage for criticizing a draft of this paper.

paragraph to the implications of computers for a factor-analytic practitioner. Although very important, I think the main thing to be said here is so obvious that we need not linger over it. With a computer one does not need to spend endless hours hanging away on a desk calculator or on punched card mechanical computers. Or, he does not need to hire, at great expense, a vast corps of clerks to carry out these efforts. Some implications of this relief from drudgery seem equally obvious. An investigator can think great thoughts about psychological and scientific problems free from inhibitions generated by purely practical computing difficulties. Also, so-called research can really no longer be based on long hours of hard work; it is much more difficult these days to set graduate students off in a corner clubbing out centroids as an excuse for their not producing a thoughtful thesis.

The other side of the coin with respect to the ease of calculations brought about by electronic computers is that investigators unquestionably are scientifically much less careful with data. Back in the Dark Ages before computers, a researcher would do some very serious psychologizing before he increased the scope of his study—when he realized that it would add perhaps weeks to the drudgery of computing; nowadays, unfortunately, he can add unrationalized variables to his matrices without blinking an eye. This seems undesirable, of course. I have also heard it said that it is undesirable that students no longer have the opportunity for that “good for the soul” plugging away—really doing the deed on a desk calculator. Things aren’t like they were in the good old days when we really had to work and appreciate our factor analyses. I consider this argument specious. As I go on shortly to more theoretical matters, I shall argue most strongly that those factor analytic computations which are desirable are impossible to do on desk calculators; only undesirable procedures may be attacked with a desk machine. It is not “good for one’s soul” by hand to invert matrices, find principal axes, and rotate analytically—things which form the very basis of factor analysis in this enlightened era; these are impossible to compute by hand.

Theory

Let me turn now to my main topic, the implications for the theory of factor analysis arising from electronic computers. It is sufficient to think of the theoretical questions of factor analysis as belonging

to two essentially independent problems. The first of these is determining an "interesting" factor space—different from the original test space. Under this are subsumed both the communality problem and the question of the number of factors. The second major problem is determining a vector basis for this "interesting" factor space. Under this are subsumed both the question of arbitrary factoring and the problem of rotation.

Communalities

In this order then, first consider the communality problem. Over the years Professor Guttman, in a series of brilliant papers, has developed the entire mathematical model of factor analysis from the scientifically and psychologically satisfying inferential viewpoint of sampling variables from a universe of psychological content. This algebraically more complicated formulation, but scientifically more insightful one, has led to a rather straightforward proof that the communality of a variable is the squared multiple correlation of this variable on the remaining variables in the universe of content. That is, the communality is the limiting squared multiple correlation on the infinite number of other variables to which we are attempting to draw psychometric inferences. Now squared multiple correlations, of course, involve inverting the correlation matrix. Additionally, Guttman essentially proved 'way back in 1940, that for the factor analytic approach to be appropriate, it is sufficient that the inverse of the correlation matrix in the limit becomes diagonal. Inverting many empirical matrices, I have found that this condition apparently is also necessary: if the inverse of the correlation matrix under consideration deviates wildly from diagonal form, subsequent operations will often lead to trouble and confusion. Thus, both these important contributions to the theory of factor analysis computationally are centered about the inverse. This is my first example of the application of computers to factor analysis. Guttman's well-appreciated, but formerly little-used ideas are now coming to light applicationally through their necessary adjunct, the ability to find results to go with the algebraic theorems with which he has provided us.

Very closely related to the mathematical model of factor analysis is a mathematical model which Guttman calls image theory or analysis. A couple of years ago it occurred to me that one of the

matrices involved in image theory represented an ideal approximation to a common-factor space. Again, we must invert the observed correlation matrix, and, since I had a computer available, I was able to try out my ideas in great quantity—something which would have been but an idle dream without a computer. Indeed, as a result of these extensive empirical efforts, I am well convinced that this image theoretical approach to the communality problem is essentially unimprovable.

Also with regard to communalities, allow me to give a small example of how computers can sometimes be an effective substitute for mathematics. Three years ago Professor Tryon and I developed what looked like an exact solution for the communality problem. This was truly exciting; we had finally found the Pot of Gold. After much arm-waving, the technique boiled down mathematically and computationally to inverting a matrix iteratively until the communalities were solved for. (Conceptually we were concerned with taking the squared multiple in the limit.) However, neither Tryon nor I was able to prove that this procedure converged to the “true” communalities. I then turned to a machine and after having computed actually four or five million squared multiple correlations, I gave up. Our approach, ultimately, was a glamorous failure. Not completely though: in rehashing why in the long pull it had failed to converge, I had impressed upon me the simple proof of why the communality problem as classically stated is insolvable.

Number of Factors

Now consider the closely interlocked problem of the number of factors. It would appear that there have been four distinguishable bases for determining the number of factors. In order of increasing importance these are (1) statistical criteria of significance, (2) algebraic criteria of necessity, (3) psychometric criteria of reliability, (4) psychological criteria of meaningfulness. Let me discuss each of these in this same order.

For those of you who have been browbeaten by the imprecations of second-rate statisticians into thinking that a significance test for the number of common factors is essential to the proper application of factor analysis, I can only say that electronic computers are an absolute necessity to carrying out such logically inappropriate statistical tests. The statistically correct but scientifically issue-con-

fusing significance tests of Lawley and Company require, I have been told, matrix inversion, and eigenvector-eigenvalue problems of the worst sort—things not ever done on anything but electronic computers.

Professor Guttman, in establishing some algebraically necessary conditions for common factor analysis, has provided us with the most important paper yet published on the number-of-factors question. More specifically, Guttman has found criteria for determining a lower bound for the number of factors. His universally strongest lower bound requires that we find the number of positive latent roots of the observed correlation matrix with squared multiples in the diagonal. An alternative lower bound—weaker than the one just mentioned—requires that we find the number of latent roots greater than one of the observed correlation matrix. I have systematically studied the first of these lower bounds through the use of a computer and have gotten results which surprised even Professor Guttman: it almost invariably is necessary—in the strict algebraic sense—to have more than half as many factors as there are variables in the study. This is not a very delightful result—considering the well-known results regarding unique communalities. I have also studied somewhat systematically Guttman's other lower bound for the number of factors, the number of eigenvalues greater than one of the observed correlation matrix. This typically runs from a sixth, say, to a third, of the number of variables.

The reason for studying this second lower bound with some care has to do with the next criterion for the number of factors, psychometric reliability. Very recently, I have worked out all of the formulas for the Kuder-Richardson reliability of factors. One remarkably simple result is that for a principal component to have positive Kuder-Richardson reliability, it is necessary and sufficient that the associated eigenvalue be greater than one—a finding corresponding exactly to Guttman's algebraic lower bound.

And, finally, from the fourth, and by far most important viewpoint for choosing the number of factors—psychological meaningfulness—I have found that the number of eigenvalues greater than one of the observed correlation matrix led to a number of factors corresponding almost invariably, in a great number of studies, to the number of factors which practicing psychologists were able to interpret.

Summarizing then, allow me to suggest a "best" answer to the question of the number of factors: it is the number of latent roots greater than one of the observed correlation matrix. This conclusion is based on the relatively independent criteria of algebraic necessity, psychometric reliability, and psychological meaningfulness. This best answer to this perennial question is not an attempt to say how many factors there are in the universe of psychological content—for the answer to this theoretical question is obviously that the number of factors in any psychological domain is infinite. I am just presenting an answer which will provide one with a basis for finding that number of factors which may be yielded necessarily, reliably, and meaningfully—from the data at hand. And this result, which you can perhaps see that I am somewhat enchanted by, has come about only through an extensive application of electronic computers; even Professor Guttman's mathematical genius is incapable of grinding out inverses and eigenvalues in the bushelsful that were necessary to reach this conclusion.

Factoring

Let us turn then to the second major problem of factor analysis: that of determining a "best" basis for the common factor space already found. This problem, of course, proceeds in two stages: first, an arbitrary factoring, and then a rotation to a purportedly "psychologically interesting" position. As for this first stage, electronic computers have provided us with a theoretically most desirable way of doing business. This is simple and not at all subtle: rather than use the vastly inferior centroid method, we may turn to the ideal procedure, Hotelling's method of principal axes. For reasons which have completely escaped me, the particularly simple distinction just made seems to have caught on slowly. Eventually, of course, because of the impact of electronic computers, the centroid method will die. From a theoretical point of view, the quicker this demise, the better.

Rotation

Once we have an arbitrary basis, then we may turn to the problem of rotation—that particular question in factor analysis for which computers relatively have probably had the greatest effect. For, before the advent of computers, rotations were always carried out

on a subjective, graphical basis. Scientifically, of course, this was nonsense, and perhaps led, more than anything else, to a bad name for factor analysis among professional mathematicians and statisticians. Amazingly, in 1953 and 1954 a number of investigators—Carroll, Neuhaus and Wrigley, Saunders, and Ferguson—independently and essentially simultaneously suddenly brought forth the first *analytic* criteria for rotation—mathematical statements through the use of to-be-optimized criterion functions of the loadings. Additionally, Professor Wrigley had the foresight to realize immediately that these ideas absolutely required the services of a computer, and, using the recently completed Illiac at the University of Illinois together with the jim-dandy name “quartimax” for his criterion, started pouring forth purely objective analytic rotational solutions by the ton. This would have been impossible without a computer. Subsequently, the development of analytic criteria for rotation has been very rapid. In 1955 I first proposed the so-called varimax criterion for analytic rotation in the orthogonal case, a criterion that—so I’ve been told—appears to be the essentially satisfactory answer for analytic orthogonal solution. Quartimax and varimax are child’s play on a computer—easily accomplished even on medium-speed machines. However, the completely general analytic solution for the case of the oblique factors has been a more difficult nut to crack. The original, and subsequently major contributor to this field, Professor Carroll, has produced a series of papers interspersed with some efforts of mine which together seem to be zeroing in on a final general solution to the rotation problem. Computationally, the oblique case is extremely difficult and time-consuming. Later in these meetings I shall give my latest attack at this problem—a possible improvement of Professor Carroll’s most recent affair—which computationally requires for a reasonably sized problem solving hundreds of latent root and vector problems of not inconsequential difficulty.

These approaches to the rotation problem—and the whole perspective which brought them about—would probably never have occurred without the atmosphere created by the potentialities of electronic computers. While computers can’t think—as the saying goes—they certainly are able to disinhibit *us* so that we may dream those voluptuous dreams which in an earlier era would have been only computational nightmares.

Main Point

I come then to what I would consider the single major point of this presentation. Factor analysis will eventually come out of the realm of a strange, mystical, ad hoc, half-art, half-science sort of numerology into the camp of reputable methodologies because of the possibility of attacking factor-analytic problems in a mathematically respectable fashion through the use of high-speed computers. Factor analysis, in the past, has been much criticized technically because of the cloacal short-cuts that have been imposed upon it by computational considerations. However, scientifically unspeakable methods of determining the number of common factors, of approximating communalities, of factoring, and of rotation can now fortunately be thought of as a dying witness to the ingenuity of earlier factor analysts in avoiding impossible computational problems.

In this vein, I might add that one of the major present deterrents to the full utilization of electronic computers in factor analysis is the error of attempting to adapt some of these antique methods to an electronic computer. For example, it strikes me as inane to use Tucker's or Coombs' technique for guessing at the number of factors now that the powerful methods of Louis Guttman are feasible. Other techniques such as using the highest in the column for a communality approximation, the centroid method, and graphical rotation, belong to an earlier era, and it is a travesty on computers and on science to continue to use them.

Programs

My next topic will be of a somewhat paternalistic sort: I shall suggest to those of you who are just building, or thinking of building, a library of programs for factor analytic purposes, some sort of priority of effort.

First, of course, the most basic program of all is that for determining intercorrelational matrices. This need not detain us, since it is an extremely easy program to write (good for beginners to get started on) and is indeed merely a data-processing problem which does not take full advantage of a high-speed electronic computer.

A second program to write, and probably the most important and most difficult, is matrix inversion for real, symmetric, positive defi-

nite matrices. An inverse program is of pre-eminent importance for a number of reasons. I have already alluded to Guttman's proof that it was sufficient for factor analytic purposes in the inferential sense that the inverse of a correlation matrix tend to be diagonal. Again referring to Guttman's work, obtaining inverses is necessary for finding the squared multiple correlations as a best approximation to the communalities, and, with respect to Guttman's work on necessary conditions, the inverse is essential for finding the universally strongest lower bound for the number of common factors. After rotating analytically, our solution, for mathematical reasons, is invariably in terms of the reference vector system of axes. It may be said that the primary factor axes are the preferable system of coordinates, and to provide the necessary transformation a matrix must be inverted. Finally an inverse program is essential for the important—and most neglected—problem in factor analysis, that of obtaining factor scores.

The third thing to program for factor analytic purposes on a computer is the principal axes method of factoring. There are, speaking very generally, two methods for doing the deed here, one due basically to Jacobi and the other to Hotelling. For factor analytic purposes Hotelling's method—involving iterated multiplication of a matrix by a vector—probably should have priority since in the arbitrary factoring of a correlation matrix we usually wish to get only a few of the largest principal axes, something Hotelling's method allows us to do. A program for finding principal axes by Jacobi's method through sine-cosine transformations should also be in the library. Jacobi's method in every respect but one seems superior to Hotelling's; it is faster per root and vector, it is less temperamental—grinding out solutions in an easily predictable time—and it is probably easier to program. Its one disadvantage is that it requires finding all roots simultaneously, something which often is not desirable in factor analysis, where we are usually interested in minimizing or maximizing a function, or getting just the dominant factors.

The fourth factor analytic procedure that I would recommend programming is the varimax criterion for analytic rotation. As indicated above, this seems to accomplish the purpose of rotating orthogonally in an essentially satisfactory manner—no one has seriously complained to me yet. It now appears that such a program for the rotation problem will be of more or less permanent usefulness.

I am somewhat at a loss to make suggestions for writing programs for analytic criteria for the general oblique case. The oblique case is presently in a state of ferment; no one has yet written a completely satisfactory analytic criterion for oblique rotation. I have hopes that the thing I shall later present at these meetings is an improvement over methods now on the market, but it is not yet what could be considered a "perfect" answer. Additionally, programs for the presently available oblique analytic criteria for rotation are very difficult to write; tentatively, I might suggest that it would be wise to delay writing such programs until the present fermentation settles down to a more permanently potable solution.

These programs I have just suggested could provide a complete system of factor analysis. For example, find all principal components with associated latent roots greater than one, rotate these according to the varimax criterion, and exactly calculate factor scores for individuals through the matrix inversion program.

Medium-Speed Computers

I should now like to turn to a topic about which I intend to speak intemperately. I should like briefly to comment on medium-speed electronic computers—of which the IBM 650 is the pre-eminent example. With the insufferable arrogance of one who has almost always dealt with high-speed computers, I should like to go on record as saying that medium-speed computers such as the IBM 650 are a positive detriment to the theoretical development of factor analysis. I consider them an anathema—machines which detract from theoretical progress while grinding out necessarily primitive solutions in immense quantities. These strong words stem from the point of view I have been taking for the last 10 or 15 minutes. You simply cannot, with these medium-speed electronic computers, do any of the things which are theoretically desirable in factor analysis—with the possible exception of rotating analytically under the restriction of orthogonality. You cannot invert large matrices on a medium-speed computer; you most certainly cannot compute principal axes of reasonably sized matrices; and you could never rotate analytically in the general oblique case except for the very smallest problems. Thus with machines of which the IBM 650 is prototypical, the methods currently used—like repulsive centroid factoring, after garbage approximation of communalities and noisome determination

of the number of factors—can, out of this slime, briefly rise to respectability through analytic criteria for rotation—as long as one maintains the sometimes undesirable restriction of orthogonality. The only conditions under which I could ever abide a medium-speed computer would be, after robbing a bank, buying my own and running it night and day just for my own problems. These incontinent words are for those of you who are contemplating procuring a medium-speed computer—or are self-satisfied with your presently completed library of programs for a medium-speed computer.

Conclusion

To conclude, allow me to repeat and rephrase my major point. In spite of the fact that computers have long been used successfully in factor analysis, the implications of electronic computers to factor analysis are still on the horizon: through their use, we will in the future more fully be able to capitalize on the before-his-time mathematical and scientific genius of Louis Guttman, the before-his-time statistical genius of Harold Hotelling, and perhaps most important, allow us to continue to translate and interpret the inspired intuitive scientific genius of L. L. Thurstone.